MZUZU UNIVERSITY

FACULTY OF INFORMATION SCIENCE AND COMMUNICATIONS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY (UPGRADING)

END OF SEMESTER EXAMINATION

ICT 4701: SOFTWARE ENGINEERING

TIME ALLOWED: 2 HOURS **DATE : JULY 12, 2016**

INSTRUCTIONS

i. This paper contains **Three** page. Please check.
ii. There are **Five questions** in this paper. Please check.
iii. Answer **ALL** questions.
iv. Marks for each question are indicated.

Question 1

a. Discuss **three** strategic options for legacy system evolution. **(6 marks)**

b. Extreme programming express user requirements as stories, with each story written on a card. Explain **two** advantages and **two** disadvantages of this approach to requirements description. **(4 marks)**

Question 2

a. What are the **four** important attributes that all professional software should have? **(4 marks)**

b. List **two** advantages and **two** disadvantages professional software engineer should be certified in the same way as doctors or lawyers. **(4 marks)**

c. List any **two** reasons why the productivity rate of programmers working as a pair is roughly the same as two programmers working individually. **(2 marks)**

d. To help counter terrorism, many countries are planning the development of computer systems that track large numbers of their citizens and their actions. Discuss the ethical implication of developing this type of system. **(2 marks)**

Question 3

a. Apart from the challenge of heterogeneity, rapid delivery and trust, identify **three** problems that Software Engineering is likely to face in the 21st century. **(6 marks)**

b. Using your knowledge of how an ATM is used, develop a set of use-cases that could serve as a basis for understanding the requirements for an ATM system. **(14 marks)**

c. Explain why the process of project planning is iterative and why a plan must be continually reviewed during a software project. **(2 marks)**

Question 4

a. What are the **five** benefits of testing software system components individually (unit and module testing) before assembling them into a complete system? **(10 marks)**

b. For large systems development, explain why it is recommended that prototypes should be throw-away prototypes. **(2 marks)**

c. Discuss **two** problems of using natural language for defining user and system requirements. **(4 marks)**

d. Discuss risk management process. **(4 marks)**

e. Discuss the following software maintenance types.

i. Corrective. **(2 marks)**

ii. Adaptive. **(2 marks)**

iii. Perfective. **(2 marks)**

iv. Preventive. **(2 marks)**

Question 5

Table 1.

Activity	Predecessor	Time Estimates			Expected Time
		Opt. (O)	Normal (M)	Pess. (P)	
A	--	2	4	6	
B	--	3	5	9	
C	A	4	5	7	
D	A	4	6	10	
E	B,C	4	5	7	
F	D	3	4	8	
G	E	3	5	8	

Key
M = Most Likely Time
O = Optimistic Time
P = Pessimistic Time

Note that duration is in days.

Refer to table 1 to answer the following questions:

a. Draw network diagram for activities in table 1. **(3 marks)**

b. Find critical path and calculate total time for project completion. **(2 marks)**

c. Calculate slack time for activities E, G, B, and D. **(4 marks)**

END OF QUESTION PAPER